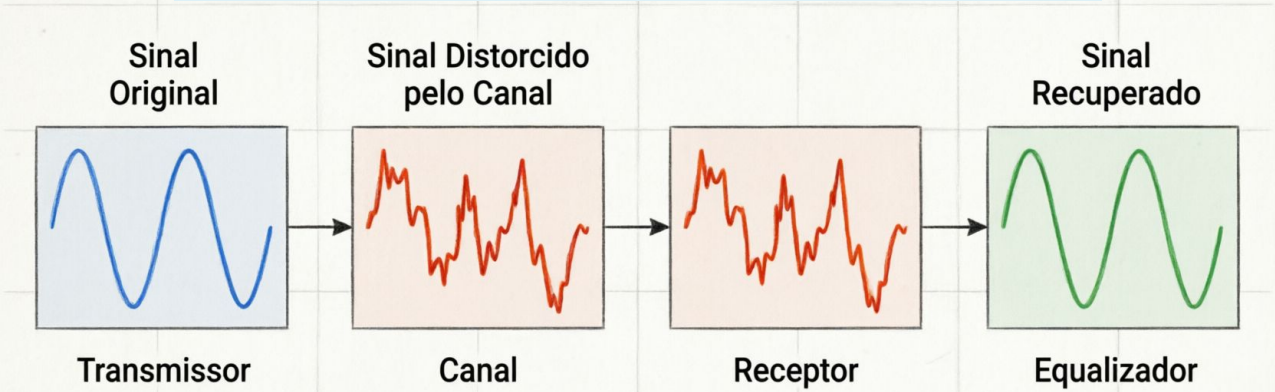
Teoria das Comunicações

Prática - Introdução à teoria das comunicações

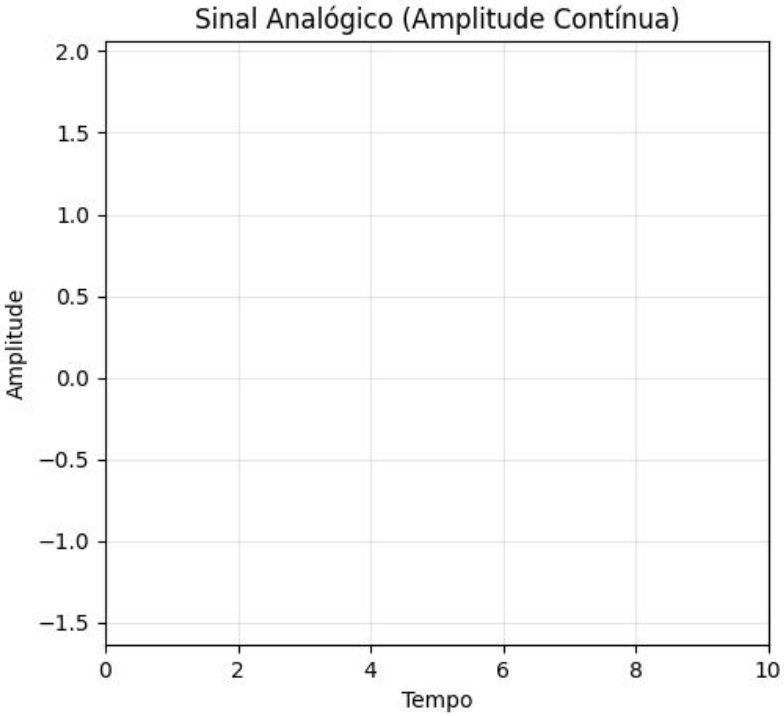
Prof. Aldebaro Klautau

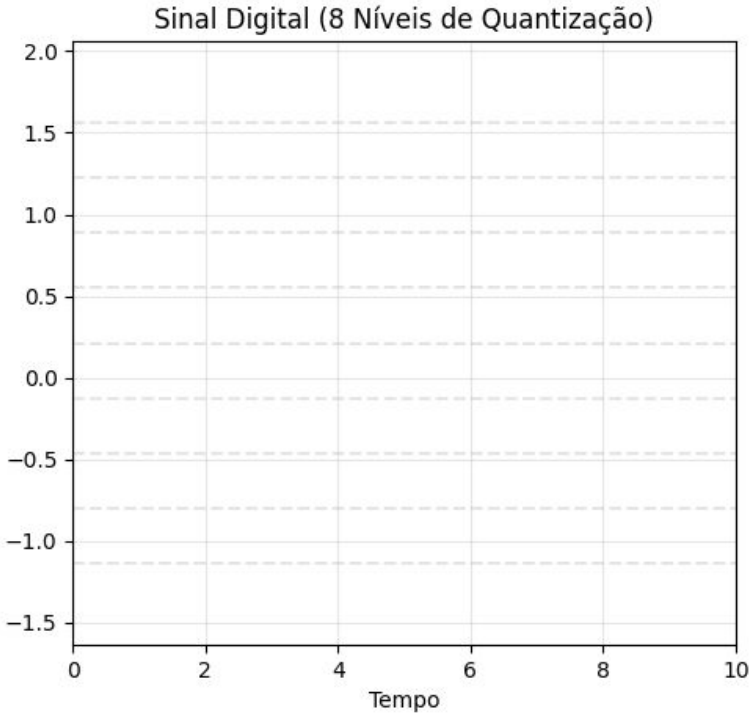
FCT/UFPA

Recapitulando - Transmissão e recepção

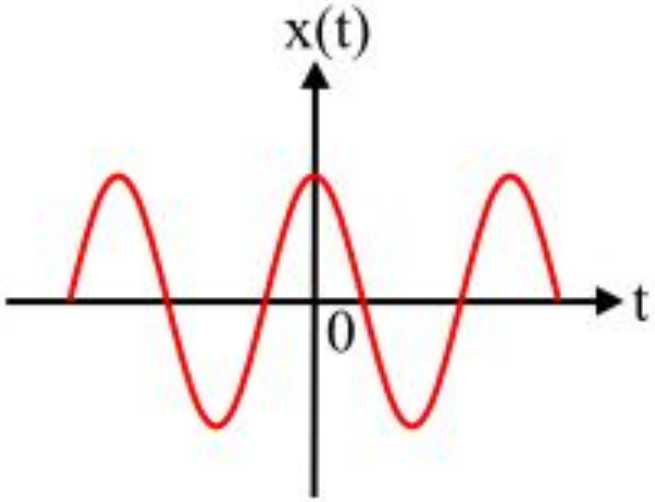


Recapitulando - Analógico x Digital

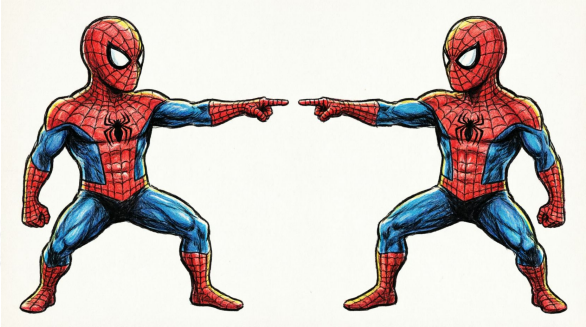


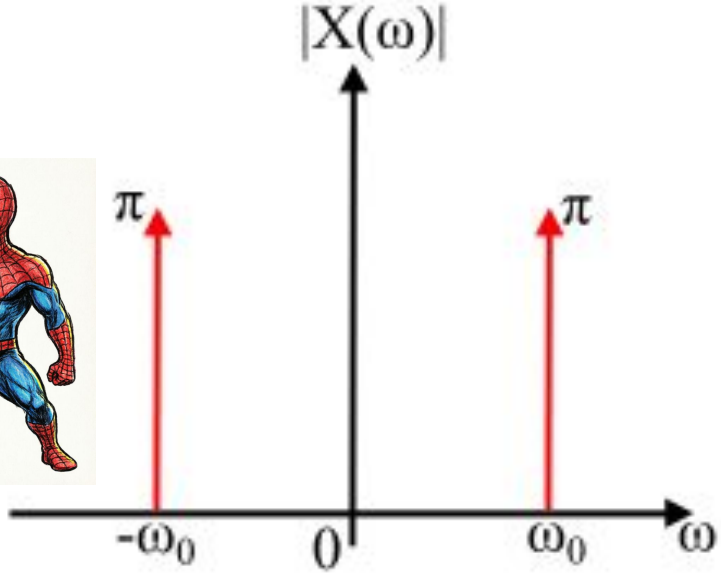


Missão da aula: Aprender sobre aplicações práticas da transformada de Fourier



Cosine Wave





Magnitude Spectrum

Figure-2

Aplicação GNU Radio

Visualizando diferentes dados usando a transformada de Fourier

A fórmula da transformada de Fourier

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi f t}\, dt$$

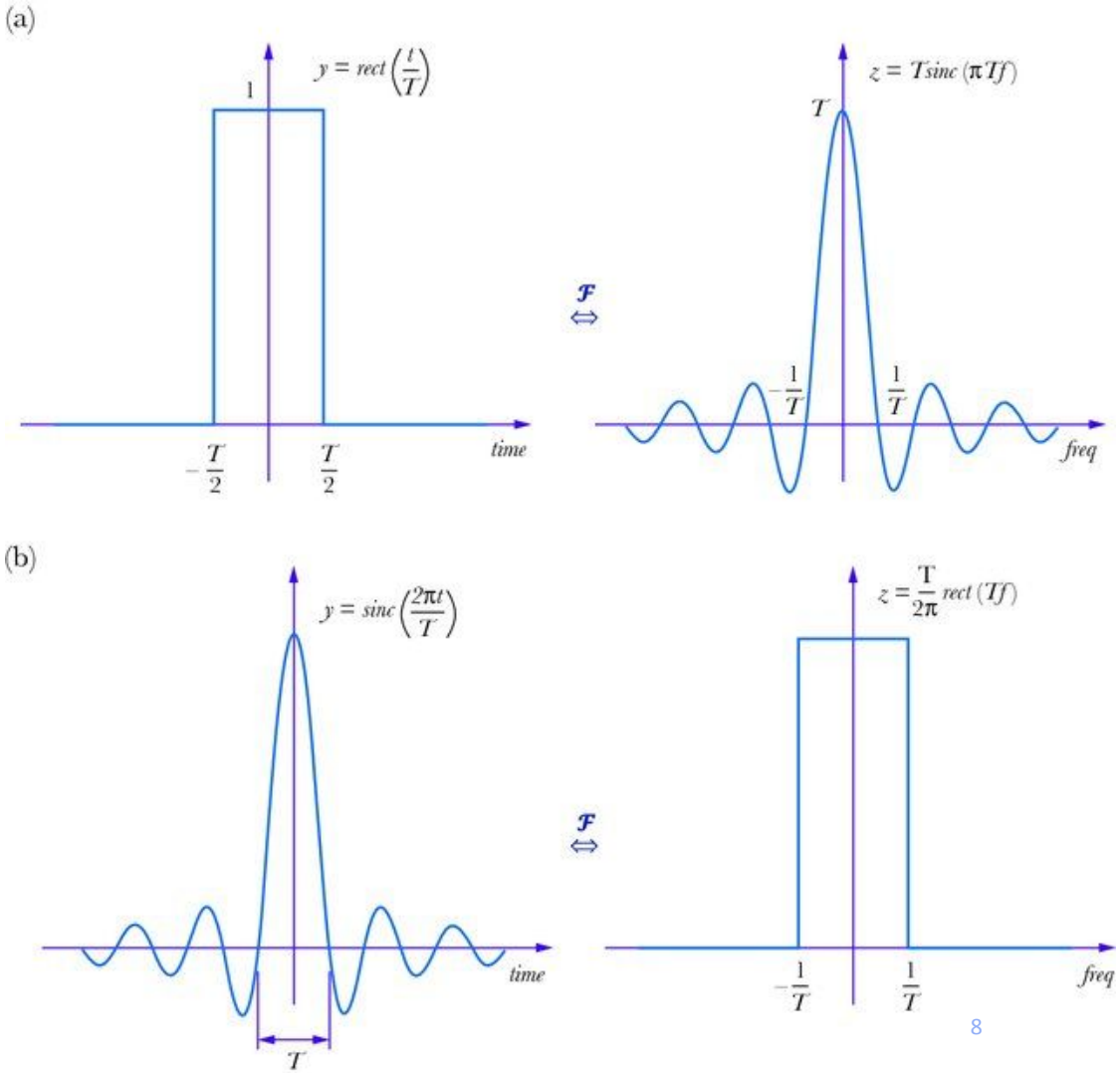
https://ai6g.org/books/dsp/FourierTransformsandSeries.html#fourier-transforms-and-series

Entendendo como a soma de senos e cossenos

$$\boxed{e^{-j\theta} = \cos(\theta) - j\sin(\theta)}$$

$$\boxed{X(f) = \int_{-\infty}^{\infty} x(t)\cos(2\pi f t)\,dt - j\int_{-\infty}^{\infty} x(t)\sin(2\pi f t)\,dt}$$

Algumas transformações úteis para o curso



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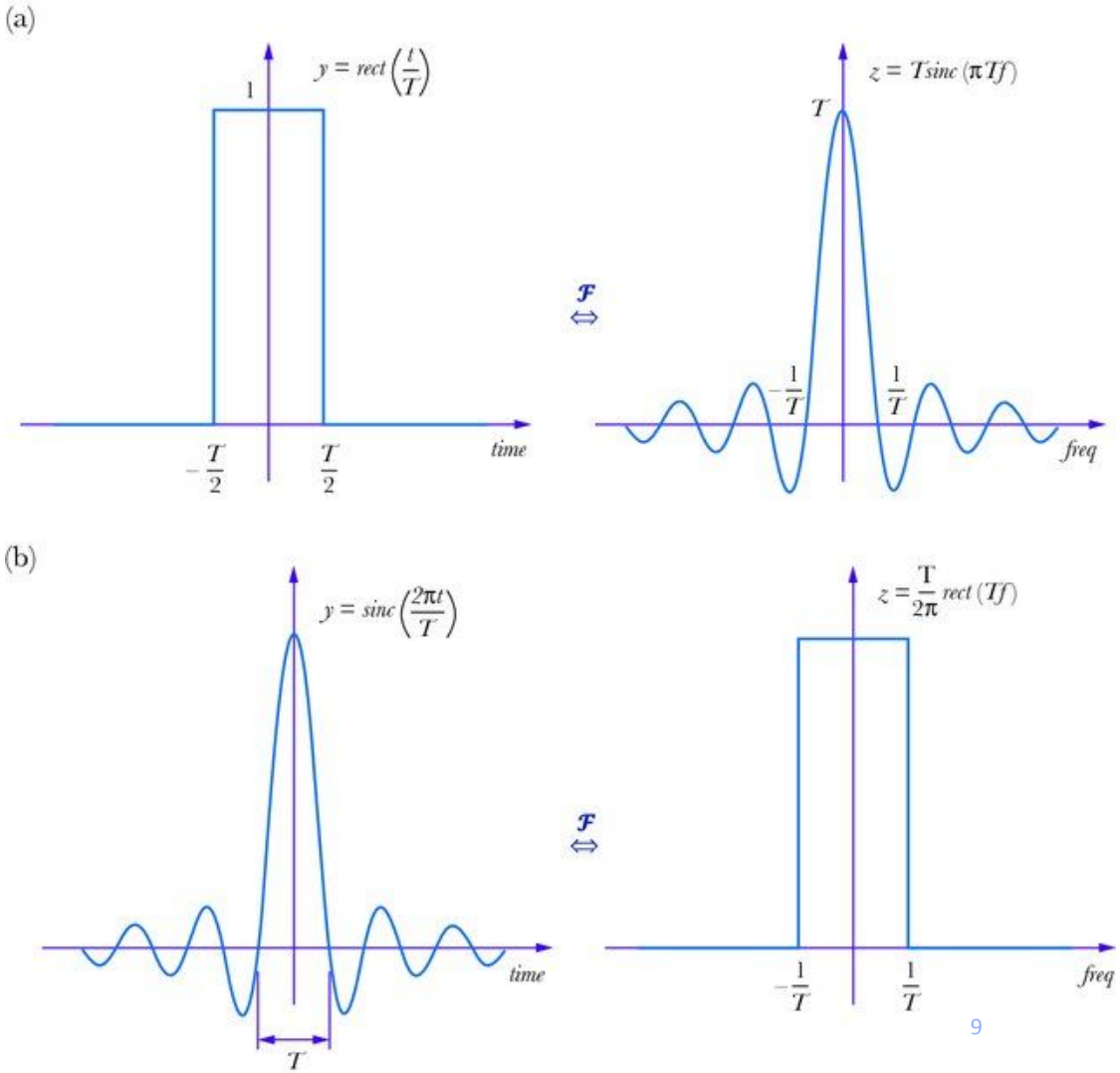

Aplicação notebook python

Refletindo sobre a representação da voz na frequência

Leituras recomendadas

- Sobre a transformada de Fourier:
  - https://ai6g.org/books/dsp/TransformsandSignalRepresentation.html#transforms-and-signal-representation
  - https://www.youtube.com/watch?v=spUNpyF58BY&pp=ygUXdHJhbnNmb3JtYWRhIGRlIGZvdXJpZXI%3D
- Sobre aprendizado de GNU Radio
  - Begginer tutorials: https://wiki.gnuradio.org/index.php/Tutorials
  - https://www.youtube.com/watch?v=q0HLlEWG8Fg&list=PLkU4tBWxFRrG_ftCf5etNE8QlhLT-alxm